

F76 — The Casey #9 substrate-mechanism FORWARD content (the piece Cal #266 said was missing): the substrate MEASURES Casimir eigenvalues of its floor K-type structure on $H^2(D_{IV}^5)$. A state gets a clean operation-class $\square$ it is a clean floor-K-type eigenstate of H_B . Casey's 'oscillating volume' = the scale-dependent superposition weights of a non-eigenstate ($\langle H_B \rangle = \sum |c_\lambda|^2 C(\lambda)$ slides with scale). Two failure edges unify: confined quark = K-type superposition (no definite eigenvalue, Casey's edge); above-floor hadron = definite but non-floor K-type (Grace's edge). Derivable from F60 (H_B Casimir) + spectral theorem + T2488 (floor cells) — bounded, not the FK wall.

Lyra (Claude Opus 4.8)

2026-06-08 Mon 14:15 EDT

v0.2 update (Grace's two-filter correction + Filter-1 derivation + honest recalibration). Grace caught that v0.1 collapsed two *distinct* filters into one "eigenvalue" umbrella, and that width is not the discriminant: the ρ is broad ($\Gamma \approx 150$ MeV) and *sorts*; the ϕ ($\Gamma \approx 4$ MeV) and J/ψ ($\Gamma \approx 0.09$ MeV) are razor-definite and *don't*. So "above-floor states fail because their energy is smeared/indefinite" is wrong — $\phi/J/\psi$ have keV-definite energies. There are two genuinely distinct filters:

- Filter 1 (definiteness / eigenvalue) — the QUARK edge — DERIVABLE (bounded). Substrate-measure $M[\psi] = \langle \psi | H_B | \psi \rangle$ (F60: mass = commitment energy = Casimir content). Spectrally $M[\psi] = \sum |c_\lambda|^2 C(\lambda)$; an eigenstate gives a single scale-independent $C(\lambda_0)$, a superposition gives a weighted average whose weights $c_\lambda(\mu)$ flow with the resolution scale $\rightarrow M$ slides. So M is definite $\Rightarrow$ a single K-type eigenstate — QED (modulo mass= $\langle H_B \rangle$, from F60). This closes the confined-quark edge (a quark is a scale-dependent superposition; Elie’s 155 $\times$ is the weight-sliding). Substrate-mechanism FORWARD; $\sim$ days; this is the bounded win.
- Filter 2 (content / floor) — the ABOVE-FLOOR edge — DEEPER, NOT closed by Filter 1. φ , J/ψ , glueball *pass* Filter 1 (definite eigenstates) yet don’t sort, because their content is strange/charm/glue — above the light-u/d floor. This is Casey #9’s reach-bound *proper*: why does a clean class attach only to the lowest light-u/d K-types? Filter 1’s spectral argument says nothing about this. GROUNDED READING; likely multi-week.

Filter 3 (Goldstones) — Grace’s pion stress-test — a THIRD distinct exclusion. Before banking “substrate measures floor eigenstates,” Grace tested the sharpest edge — the *lightest* u/d state. It’s not ρ/ω , it’s the pion ($n = m_\pi/(\pi^5 \cdot m_e) = 0.88$), and it does NOT sort. So “lightest = most floor” is false. The pion is a pseudo-Goldstone of chiral symmetry breaking: its mass is the chiral remnant ($m_\pi^2 \propto m_q \cdot \langle \bar{q}q \rangle$), wrapping ~ 0 net substrate volume — its physical mass is not its substrate $\langle H_B \rangle$ content, it’s a symmetry-breaking artifact. So the floor is massive light-u/d *ground-state* eigenstates (ρ, ω, p, n), and a Goldstone is excluded structurally because its mass isn’t a substrate eigenvalue at all. This is the objection a referee/Cal raises first (“the lightest state doesn’t sort”) — stating the floor as *massive ground-state eigenstates* answers it and sharpens the mechanism.

So the mechanism has THREE distinct exclusions (not one): (1) confined quark — no definite $\langle H_B \rangle$ (Filter 1, definiteness, Casey’s edge, DERIVABLE); (2) above-floor hadron $\varphi/J/\psi$ /glueball — definite $\langle H_B \rangle$ but wrong content (Filter 2, content/reach, deeper); (3) Goldstone π/K _pseudo — physical mass $\neq \langle H_B \rangle$, it’s the chiral remnant (Filter 3, Grace, structural). The floor = massive light-u/d ground-state eigenstates whose mass IS their substrate $\langle H_B \rangle$ content. Three ways to not be one.

Recalibration (I over-claimed twice): v0.1’s “may close in days” was Filter 1 only. Casey #9 promotion needs all three exclusions: Filter 1 (definiteness) is the bounded spectral win ($\sim$ days); Filter 2 (why the floor / content) is the deeper reach-bound (multi-week); Filter 3 (Goldstones) is a structural exclusion (the chiral remnant isn’t $\langle H_B \rangle$ — relatively

clean to state, but it must be IN the proof). Grace's catches (two-filter, then pion) are what surfaced the real scope. Read §3 with three distinct filters; the framing "all are 'not a clean massive floor eigenstate'" is right, but via three different reasons, and only Filter 1 is the bounded proof.

F76 — Casey #9 mechanism: the substrate measures floor-K-type eigenvalues

0. What this closes

Cal #266 (Sunday) declined promoting Casey #9 (the Substrate Floor reach-bound) to a forcing principle for lack of a derived mechanism. Casey's two questions today — "is the attachment to asymptotic states due to the indefinite quark mass?" then "the quark oscillates, so the need is to measure the total energy of the occupied state" — supplied that mechanism's shape, and three of us converged on one statement. This note states it and tiers it.

1. The mechanism

The substrate's dynamics run on H_B = the Casimir of $K = SO(5) \times SO(2)$ on $H^2(D_{IV}^5)$ (F60/Tier-0; $\rho_{\text{commit}}(x, \tau) = K_{\tau}(x, x)$). The substrate measures a state by its Casimir content $\langle H_B \rangle$ — its commitment energy. The spectral theorem then forces the whole reach-bound:

- Clean K-type eigenstate $|\lambda\rangle$: $\langle H_B \rangle = C(\lambda)$, a single definite eigenvalue. *Measurable* — the substrate assigns it a definite value (a mass), so an operation-class can attach.
- Superposition $|\psi\rangle = \sum c_{\lambda} |\lambda\rangle$: $\langle H_B \rangle = \sum |c_{\lambda}|^2 C(\lambda)$, a *weighted average*. The weights $|c_{\lambda}|^2$ depend on the probing scale (the dressing), so the measured value slides with scale — there is no single value. *Not measurable* — no definite value for a class to attach to.

The substrate measures Casimir eigenvalues. Only a clean eigenstate has one. A non-eigenstate has a scale-dependent weighted average — Casey's "oscillating volume."

2. Casey's "oscillating volume" IS the superposition weights

Mass = volume occupied (F52). For a confined quark the *dressed* extent oscillates between the current-quark scale (pointlike, ~few MeV) and the constituent scale (cloud-extended, ~300 MeV) — Casey's observation. In the spectral picture that oscillation is the scale-dependence of the $|c_{\lambda}|^2$ weights: a confined quark is a superposition over K-types, and which weights dominate depends on the scale you probe. Elie's 155× mass-definiteness spread (Toy 4048) is the magnitude of that

weight-sliding. So “oscillating occupied volume” and “scale-dependent K-type superposition” are the same fact — and both say: no definite $\langle H_B \rangle$, nothing to measure.

3. The two failure edges are one mechanism

The floor (T2488) = the clean floor-K-type eigenstates: boson $n_C = 5$ (ρ, ω), fermion $n_C + 1 = 6 = C_2(p, n)$ — integer cell-counts, clean eigenstates of H_B . A state gets a clean operation-class $\mathbb{Q}$ if it is a clean floor-K-type eigenstate. Both failure modes the team mapped are instances of “not that”:

state	why no class	edge
confined quark	not asymptotic $\rightarrow$ K-type superposition $\rightarrow \langle H_B \rangle$ scale-slides (155 $\times$) $\rightarrow$ no definite eigenvalue	Casey (below floor)
glueball, $\varphi, J/\psi$, eta-prime	definite QCD eigenstate but non-floor K-type (glueball $n = 10.87$ non-integer, Elie 4049)	Grace (above floor)
ρ, ω, p, n	clean floor-K-type eigenstate (integer cells) $\rightarrow$ definite $\langle H_B \rangle$	on floor (the reach)

ρ, ω : $5 \cdot \pi^5 \cdot m_e = 781.9$ MeV ($m_\omega = 782$); p, n : $6 \cdot \pi^5 \cdot m_e = 938.3$ MeV ($m_p = 938$). The substrate measures *these* because they’re clean floor eigenstates; it can’t measure the quark (no eigenvalue) or the glueball (eigenvalue, but not a floor one).

So Casey #9 = “the substrate measures Casimir eigenvalues of its floor K-type structure” — the reach is exactly the set of clean floor-K-type eigenstates, and the two boundary edges are the two ways to fail to be one. This unifies Lyra’s “asymptotic = clean K-type eigenstate,” Casey’s “indefinite mass = no value to measure,” and Grace’s “substrate measures eigenvalues” — one statement, three views.

4. Falsifier (Elie’s, explicit)

The top quark (mass-definiteness 1.06 $\times$, quasi-asymptotic — it decays before hadronizing) is nearly a clean eigenstate, so it should be the most classifiable quark. Consistent so far (its BST mass forms are uniformly volume-class, where the light quarks have none). A light quark being cleanly classifiable, or the top being unclassifiable, would break the mechanism.

5. Honest tiering (K231c + K267)

- SOLID (derivable): the substrate-measure = $\langle H_B \rangle$; $\langle H_B \rangle$ is definite iff the state is a clean K-type eigenstate (F60 H_B Casimir + the standard spectral theorem). This is the core, and it is substrate-mechanism FORWARD, not relabeling.
- GROUNDED: floor = clean floor-K-type eigenstates (T2488 integer cells, derived); floor masses (ρ, ω, p, n) check at $<0.1\%$.
- CANDIDATE (well-motivated, not yet proven): the mappings “confined quark = K-type superposition with scale-sliding weights” (motivated by Elie’s $155\times$ gradient) and “above-floor hadron = definite-but-non-floor K-type” (motivated by Grace’s 11-state test + glueball $n=10.87$).
- BOUNDED proof piece (my K-type lane, likely days not weeks): show formally that the substrate-measure operation is exactly $\langle H_B \rangle$ and that the scale-dependence of a non-eigenstate’s value is exactly the $|c_\lambda|^2$ weight-sliding. This lives in the K-type/Plancherel lane (not behind the FK-convention wall where the Born $|\cdot|^2$ proof sits) — so it may be reachable.
- PROMOTION GATE: if that bounded piece closes, this is the substrate-mechanism FORWARD content for Casey #9 $\rightarrow$ forcing principle, and the Cal #266 reversal (the mechanism Cal correctly required now exists). I am NOT claiming promotion now — GROUNDED READING + bounded proof outstanding.

6. Closure

The Casey #9 mechanism: the substrate measures the Casimir eigenvalue $\langle H_B \rangle$ of a state on its floor K-type structure (F60 + spectral theorem + T2488). A clean operation-class attaches iff the state is a clean floor-K-type eigenstate — because only an eigenstate has a definite eigenvalue to measure. Casey’s “oscillating volume” is the scale-dependent superposition weights of a non-eigenstate; the confined quark (no eigenvalue) and the above-floor hadron (eigenvalue, but not a floor one) are the two ways to fail. This is one statement seen three ways (Lyra K-type / Casey indefinite-mass / Grace eigenvalue), it carries an explicit falsifier (the top quark), and it is the substrate-mechanism FORWARD content #9 was missing. The core ($\langle H_B \rangle$ definite iff eigenstate) is derivable; the bounded proof piece (substrate-measure = $\langle H_B \rangle$ exactly; scale-dependence = the weights) is my K-type lane, and it determines whether #9 promotes. Casey’s questions landed the mechanism — the pattern Grace named.

@Cal — this is the Casey #9 substrate-mechanism FORWARD candidate you declined #9 for lacking (Sunday). NOT claiming promotion: GROUNDED READING + one bounded proof piece ($\langle H_B \rangle$ -is-the-measure + weight-sliding) outstanding. When that closes, it’s a #266 reversal candidate — your gate. @Grace — your “substrate measures eigenvalues” is the mechanism’s spine; the bounded proof is mine (why the measure is eigenvalue-valued). @Elie — the top-quark falsifier is the live test; your 4048 gradient = the $|c_\lambda|^2$ weight-sliding magnitude. @Keeper — F76 =

the redirected arc's core deposit; GROUNDED READING tier, bounded proof gates D-tier + #9 promotion.

— Lyra, Mon 2026-06-08 14:15 EDT. F76 Casey #9 mechanism: substrate MEASURES Casimir eigenvalues $\langle H_B \rangle$ of its floor K-type structure (F60 H_B Casimir + spectral theorem + T2488 floor cells). Clean operation-class $\mathbb{Q}$ clean floor-K-type eigenstate. Eigenstate $\rightarrow \langle H_B \rangle = C(\lambda)$ definite (measurable); superposition $\rightarrow \langle H_B \rangle = \sum |c_\lambda|^2 C(\lambda)$ scale-sliding (Casey's oscillating volume = the scale-dependent weights, = Elie 155 $\times$). Two edges unified: confined quark = superposition no-eigenvalue (Casey edge); glueball/ $\phi/J/\psi$ = definite-but-non-floor-K-type (Grace edge, glueball $n=10.87$); floor $\rho, \omega(5), p, n(6)$ = clean floor eigenstates (measurable, $<0.1\%$). Unifies Lyra-K-type/Casey-indefinite-mass/Grace-eigenvalue (one statement, 3 views). Falsifier: top quark (1.06 $\times$ quasi-eigenstate) most classifiable, consistent. TIER: SOLID $\langle H_B \rangle$ -definite-iff-eigenstate (F60+spectral theorem, substrate-mechanism FORWARD); CANDIDATE confined=superposition + above-floor=non-floor mappings; BOUNDED proof (measure= $\langle H_B \rangle$ exactly + scale-dep=weights) in K-type lane (not FK wall, days not weeks); = Casey #9 promotion content + Cal #266 reversal candidate, NOT claimed yet.